Cover letter

Dear Editor,

Please consider the revised manuscript, “Candidate Composition, Ancestor Scores, and Local Mixing in Public Patent Topic Ranking,” by Haijie Zhang and Weidong Liu. We suggest the Research Article category. The paper combines positive and negative controlled findings, with its main contribution in the evaluation of classification-derived patent ranking.

The manuscript now uses a traceable public HUPD cohort of 18,711 analytical inputs. The revised target is primary-IPC topic-tree agreement within supplied forty-candidate lists. The experiments distinguish candidate composition, access to tree topology and finite-budget feature transformations, and local mixing objectives. The accompanying response explains the expanded capacity and tuning controls, exact list inventories, grade-boundary analyses and reconstruction materials. It distinguishes the original five repeated refits from secondary one-seed extensions.

We disclose the related published conference paper by Weidong Liu and Haijie Zhang, “Innovation Hierarchy Based Patent Representation Model,” IJCNN 2022, DOI 10.1109/IJCNN55064.2022.9892513. The current study shares the patent-representation motivation and topic, but does not use that paper's nested hierarchical Dirichlet process or variational inference. Its public HUPD cohort, classification-derived evaluation, controlled comparisons and numerical results are separately specified. The prior paper is cited in the manuscript. The earlier confidential-data manuscript has not been published and is not under consideration at another journal, as confirmed by the author. Its confidential data and outcomes are not evidence for the current public experiment.

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The project address supplied by the authors is https://github.com/MrHJie/technical-text-ranking. The public tree inspected on October 31, 2026 contained an earlier pipeline. The HUPD revision artifacts are provided in the accompanying package, with hashes and instructions for result reconstruction, exact score replay and fresh fitting. We do not claim an uncreated DOI or a completed public deposit of this package. The original HUPD source is independently public and pinned by revision and hashes.

Thank you for considering the revised work.

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